

THE EFFECT OF CARFEINE URON BLOOD GLUCOSE LEVEL

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Diabetic patients often drink genuine coffee to reduce fatigue and the frequently associated "physical exhaustion symptoms." The general improvement in feeling observed after the drinking of coffee is related to circulatory changes in the system [2, 9, 11] caused primarily by the caffeine contained in the coffee [12, 13].

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The pharmacology of caffeine is well-known; however, the mechanism of its biochemical effect still has not been explained conclusively. According to Hauschild [5], the most convincing view is that of Fröhlich and Zak [3]. In the opinion of these authors, caffeine increases the permeability of the cell membrane in the capillaries, and also the intracellular metabolism, which in turn leads to dilation of the vessels. Reduced permeability of the cell membrane is generally considered as one of the most pathogenic factors in diabetes, responsible for slowing down processes which break down the glucose in the cells and increase its level in the blood [1, 4, 6, 10]. If we assume that caffeine changes the permeability of the cell membrane not only in the endothelia of the blood vessels but also in other tissues, its role and effect on the system, especially among diabetic patients, will be seen in a new light which interests us. The idea in these assumptions is that caffeine, like insulin, will improve the efficiency of the glucose processes in the cells and thus reduce the glucose level in the blood.

* Numbers in the margin indicate pagination in the foreign text.

I attempted to demonstrate the validity of these theoretical assumptions among patients, especially patients suffering from diabetes. An additional incentive for carrying out these studies was also the well-known observations that among people not used to genuine coffee, symptoms similar to those observed in moderate hyperglycemia occurred after the coffee was drunk (a tendency to perspire, "internal restlessness" and, sometimes, even tremors).

Observations and Method

Adult outpatients, whose blood sugar level was determined in the laboratory, were examined. Persons whose state of health made the use of caffeine inadvisable (persons with high blood pressure and persons sensitive to chemicals and medications) were excluded from the study. The caffeine was administered in three forms and doses: subcutaneous injection of caffeine sodium benzoate, manufactured by the Polish Pharmaceutical Plants (on the average, 0.08-0.1 g pure caffeine), orally, in the form of genuine coffee (from "Delicacies") in the amount of 5 g/100 ml boiling water (according to Walewski and Skrodzki [14], this corresponds to a 0.05-0.08 g caffeine dose), and in the form of a 5% aqueous pure caffeine solution (KHF packing) (0.15 g caffeine/30 ml).

A 0.1 ml blood sample was taken from the finger before the study and at the appropriate time after administration of the caffeine. According to Seyffert [11], caffeine administered orally, i.e. in the form of coffee or a solution, begins to have an effect after 10-20 min, and the strongest effect occurs after 1-2 hours. Obviously, this time is much shorter when the caffeine is injected. However, due to the outpatient character of the studies (the limited time of the patients), I was forced to take the second blood sample after a shorter time, namely 30-35 min after injection of the caffeine sodium benzoate and 45-50 min after oral administration of the coffee or caffeine solution. I

used the Hagedorn-Jensen method to determine the glucose level in the blood.

TABLE 1

	Number of persons examined	Including	
		Women	Men
Caffeine sodium benzoate	101	89	12
Genuine coffee	54	35	19
Caffeine solution	73	45	28
Total	228	169	59

In the cases that were observed, in addition to the glucose level in the blood, I took into account other factors such as the age of the patients, the duration of the disease, the method used to treat the diabetes (among diabetic patients), and also whether they drank or did not drink genuine coffee. These data enabled me to analyze the results of the study more thoroughly.

Results

Prior to the actual studies, I determined the glucose level in the blood 100 times from one sample. The mean error of measurement was $(\delta) = \pm 8 \text{ mg-\%}$. To facilitate the study, the measurement error was taken as $\pm 10 \text{ mg-\%}$ (this corresponds to 1.258), thanks to which the results could be evaluated more accurately [8]. The numerical values $X - X_0 = \pm X$, which, depending on the sign, corresponded to a decrease (-), increase (+), or no change (0) in the glucose level in the blood, were obtained from measurements of the differences in the glucose level after the administration of the caffeine (X) and before its administration (X_0). All patients were classified into groups. The glucose level

in the blood before caffeine administration was used as the basis for the classification. Group I included subjects in whom a normal glucose level was determined (healthy persons), group II, patients with severe hyperglycemia (above 300 mg-%), group III, patients with moderate hyperglycemia (200-300 mg-%), and group IV, persons with slight hyperglycemia (below 200 mg-%).

Discussion of Results

The decrease in the glucose level in the blood after administration of the caffeine was significant -- on the average, 12.1 mg-% among all persons studied (228 people), 11.6 mg-% among the healthy persons (52 people), and 13.3 mg-% among diabetic patients (176 people). The difference in the decrease in the level among healthy people and diabetic patients is not large when we consider the actual values without taking into account the age of the persons studied and the degree of hyperglycemia. A more pronounced change in the glucose level was detected among younger persons of whom, in my study, a large number was in the group of healthy persons. Among diabetic patients (Fig. 1), the decrease in the glucose level was most pronounced in the group of persons with hyperglycemia above 300 mg-% (30.7 mg-%) (i.e. about 10% of the original value), and smallest in the group with slight hyperglycemia (up to 200 mg-%), where it was 5.8 mg-% (about 2.5% of the original value). The correlation between the degree of glycemia and the reduced glucose level value obtained after administration of the caffeine was studied statistically (Table 2).

When all persons in whom the glucose level decreased after the administration of caffeine were taken into account (N = 111 persons examined), the statistical significance of the sample correlation coefficient demonstrated that the reduction in the value depends on the degree of glycemia. On the other hand, the value of the regression coefficient b (for the